Trigonometric Functions & the Unit Circle

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Radian Measure & Unit Circle Coordinates**

A terminal ray in standard position passes through the point $P(-8, 15)$. Find the exact values of $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Step 1: Compute the radial distance $r = \sqrt{x^2 + y^2}$: $r = \sqrt{(-8)^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$.
Step 2: Apply circular trigonometric definitions:
$\sin\boldsymbol{\theta} = \frac{y}{r} = \frac{15}{17}$, $\cos\boldsymbol{\theta} = \frac{x}{r} = -\frac{8}{17}$, $\tan\boldsymbol{\theta} = \frac{y}{x} = -\frac{15}{8}$.

2 **[Foundational]** • **Target: Exact Values of Trigonometric Functions**

Given vectors $\mathbf{u} = \langle 4, -3\rangle$ and $\mathbf{v} = \langle -1, 5\rangle$, compute: (a) $2\mathbf{u} - 3\mathbf{v}$, (b) the magnitude $||\mathbf{u}||$ and unit vector in the direction of $\mathbf{u}$, and (c) the dot product $\mathbf{u} \cdot \mathbf{v}$.

(a) Linear combination: $2\langle 4, -3\rangle - 3\langle -1, 5\rangle = \langle 8 + 3, -6 - 15\rangle = \boldsymbol{\langle 11, -21\rangle}$. (b) Magnitude and unit vector: $||\mathbf{u}|| = \sqrt{4^2 + (-3)^2} = \mathbf{5}$, $\hat{\mathbf{u}} = \left\langle \frac{4}{5}, -\frac{3}{5} \right\rangle$. (c) Dot product:
$\mathbf{u} \cdot \mathbf{v} = (4)(-1) + (-3)(5) = -4 - 15 = \mathbf{-19}$.

3 **[Foundational]** • **Target: Sinusoidal Graphs & Transformations**

State the amplitude, period, phase shift, and midline of $f(x) = 3\sin(2x - \pi) + 1$.

Step 1: Factor the argument into standard sinusoidal form $f(x) = A\sin(B(x - C)) + D$:
$f(x) = 3\sin\left(2\left(x - \frac{\pi}{2}\right)\right) + 1$. Step 2: Read off the four sinusoidal invariants:Amplitude: $|A| = |3| = \mathbf{3}$
.Period: $T = \frac{2\pi}{|B|} = \frac{2\pi}{2} = \boldsymbol{\pi}$.Phase Shift: $C = \boldsymbol{\frac{\pi}{2}}$ units to the right.Midline: Horizontal line $\mathbf{y = 1}$.

4 **[Foundational]** • **Target: Fundamental Trigonometric Identities**

A population doubles every 12 years, starting from $P_0 = 500$. When does it reach 4000?

Step 1: Formulate the exponential doubling model: $P(t) = P_0 \cdot 2^{t/12} = 500 \cdot 2^{t/12}$. Step 2: Set $P(t) = 4000$ and isolate the exponential base: $500 \cdot 2^{t/12} = 4000 \Rightarrow 2^{t/12} = \frac{4000}{500} = 8$. Step 3: Equate exponents of base 2: $2^{t/12} = 2^3 \Rightarrow \frac{t}{12} = 3 \Rightarrow t = \mathbf{36}$ years.

5 **[Foundational]** • **Target: Radian Measure & Unit Circle Coordinates**

Convert the polar equation $r = 6\cos\theta - 8\sin\theta$ to Cartesian rectangular form, and identify the geometric curve, its center, and its radius.

Step 1: Multiply both sides by r: $r^2 = 6r\cos\theta - 8r\sin\theta$. Step 2: Substitute $r^2 = x^2 + y^2$, $r\cos\theta = x$, and $r\sin\theta = y$: $x^2 + y^2 = 6x - 8y \Rightarrow (x^2 - 6x + 9) + (y^2 + 8y + 16) = 9 + 16 = 25$.
$\boldsymbol{(x - 3)^2 + (y + 4)^2 = 25}$. The curve is a circle with center $\boldsymbol{(3, -4)}$ and radius $\mathbf{R = 5}$.

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6 [Foundational] • Target: Exact Values of Trigonometric Functions

Eliminate the parameter t from the parametric curve $x(t) = 3 + 4\cos t$, $y(t) = -2 + 5\sin t$ for $0 \le t \le 2\pi$. Identify the conic section, its center, and its vertices.

Step 1: Isolate $\cos t$ and $\sin t$: $\cos t = \frac{x-3}{4}$, $\sin t = \frac{y+2}{5}$. Step 2: Apply the Pythagorean identity $\cos^2 t + \sin^2 t = 1$: $\mathbf{\frac{(x-3)^2}{16} + \frac{(y+2)^2}{25} = 1}$. This is a vertical ellipse centered at $\mathbf{(3, -2)}$ with major-axis vertices at $\mathbf{(3, -7)}$ and $\mathbf{(3, 3)}$.

7 [Foundational] • Target: Sinusoidal Graphs & Transformations

A particle moves along $x(t) = 2t + 1$, $y(t) = t^2 - 3$. Find the Cartesian equation of the path.

Step 1: Solve the horizontal parametric equation for parameter t: $x = 2t + 1 \Rightarrow 2t = x - 1 \Rightarrow t = \frac{x-1}{2}$. Step 2: Substitute t into the vertical equation $y(t)$: $y = \left(\frac{x-1}{2}\right)^2 - 3 = \mathbf{\frac{(x-1)^2}{4} - 3}$. The trajectory is an upward-opening parabola with vertex at $(1, -3)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Fundamental Trigonometric Identities

A transformation maps point $A = (1, 2)$ to $A' = (-4, 2)$. If this transformation is a translation, what is the component form of the translation vector $\mathbf{v}$?

Let the initial point be $A = (x_1, y_1)$ and the translated point be $A' = (x_2, y_2)$. The translation vector $\mathbf{v}$ is given by $\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, $(x_1, y_1) = (1, 2)$ and $(x_2, y_2) = (-4, 2)$. Therefore, $\mathbf{v} = \langle -4 - 1, 2 - 2 \rangle = \langle -5, 0 \rangle$.

9 [Intermediate] • Target: Radian Measure & Unit Circle Coordinates

Consider the transformation grid below. If the original point A is mapped to A', what is the vector representing this translation?

The vector representing a translation from point $P = (x_1, y_1)$ to point $P' = (x_2, y_2)$ is given by $\langle x_2 - x_1, y_2 - y_1 \rangle$. In this case, point A is at $(1, 2)$ and point A' is at $(-5, 2)$. Therefore, the translation vector is $\langle -5 - 1, 2 - 2 \rangle = \langle -6, 0 \rangle$.

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10 [Intermediate] • Target: Exact Values of Trigonometric Functions

A point P is on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. The coordinates of P are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. If a point Q on the unit circle has an angle ϕ such that $\phi = \theta + \frac{\pi}{3}$, what is the y-coordinate of point Q? Show your complete algebraic work.

Final Answer: $\frac{\sqrt{3}}{2}$. The coordinates of point P on the unit circle are given as $(\cos\theta, \sin\theta)$. Therefore, $\cos\theta = \frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$. This implies that $\theta = \frac{\pi}{3}$ (or 60 degrees). The point Q on the unit circle has an angle $\phi = \theta + \frac{\pi}{3}$. Substituting the value of θ, we get $\phi = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$ (or 120 degrees). The coordinates of Q are $(\cos\phi, \sin\phi)$. We need to find the y-coordinate of Q, which is $\sin\phi$. Therefore, the y-coordinate of Q is $\sin\left(\frac{2\pi}{3}\right)$. Using the unit circle or trigonometric identities, $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

11 [Intermediate] • Target: Sinusoidal Graphs & Transformations

The point P is located on the unit circle at an angle of $\frac{\pi}{3}$ radians counterclockwise from the positive x-axis. What are the Cartesian coordinates (x, y) of point P? Show your complete algebraic work.

Final Answer: $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. The unit circle is defined by the equation $x^2 + y^2 = 1$. For a point on the unit circle at an angle θ from the positive x-axis, the Cartesian coordinates are given by $x = \cos(\theta)$ and $y = \sin(\theta)$. In this case, the angle is $\theta = \frac{\pi}{3}$ radians. We know that $\cos(\frac{\pi}{3}) = \frac{1}{2}$ and $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. Therefore, the Cartesian coordinates of point P are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

12 [Intermediate] • Target: Fundamental Trigonometric Identities

A point P is located on the unit circle such that its angle with the positive x-axis is $\frac{2\pi}{3}$ radians. What are the coordinates of point P? Show your complete algebraic work.

Final Answer: $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$. The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{2\pi}{3}$ radians. We know that $\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$ and $\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$. Therefore, the coordinates of point P are $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

13 [Intermediate] • Target: Radian Measure & Unit Circle Coordinates

A circle is centered at the origin, and a point P lies on the circle. The angle formed by the positive x-axis and the line segment OP (where O is the origin) is θ. The coordinates of point P are $(r\cos\theta, r\sin\theta)$. If point P has coordinates $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$ and lies on a circle of radius R, find the value of R.

Final Answer: 5. The coordinates of a point P on a circle of radius R centered at the origin are given by $(R\cos\theta, R\sin\theta)$. We are given that the coordinates of P are $\left(\frac{5}{2}, \frac{5\sqrt{3}}{2}\right)$. Therefore, we can set up the following equations: $R\cos\theta = \frac{5}{2}$ $R\sin\theta = \frac{5\sqrt{3}}{2}$ To find R, we can square both equations and add them together, using the identity $\cos^2\theta + \sin^2\theta = 1$: $(R\cos\theta)^2 + (R\sin\theta)^2 = \left(\frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2$ $R^2\cos^2\theta + R^2\sin^2\theta = \frac{25}{4} + \frac{25\cdot 3}{4}$ $R^2(\cos^2\theta + \sin^2\theta) = \frac{25}{4} + \frac{75}{4}$ $R^2(1) = \frac{100}{4}$ $R^2 = 25$ Since R represents a radius, it must be positive. Therefore, $R = \sqrt{25} = 5$.

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14 [Intermediate] • Target: Exact Values of Trigonometric Functions

A point P lies on the unit circle such that its terminal ray makes an angle of θ with the positive x-axis. If the x-coordinate of P is $\frac{1}{2}$, what is the y-coordinate of P? Assume θ is in the first quadrant.

Final Answer: $\frac{\sqrt{3}}{2}$. For a point P on the unit circle, its coordinates are given by $(\cos\theta, \sin\theta)$. We are given that the x-coordinate of P is $\frac{1}{2}$, so $\cos\theta = \frac{1}{2}$. Since P is on the unit circle, its coordinates satisfy the equation $x^2 + y^2 = 1$. Substituting the x-coordinate, we get $(\frac{1}{2})^2 + y^2 = 1$. This simplifies to $\frac{1}{4} + y^2 = 1$. Subtracting $\frac{1}{4}$ from both sides gives $y^2 = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get $y = \pm\frac{\sqrt{3}}{2}$. Since the problem states that θ is in the first quadrant, the y-coordinate must be positive. Therefore, the y-coordinate of P is $\frac{\sqrt{3}}{2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Sinusoidal Graphs & Transformations • Cross-Lesson: 1.02 Rational Functions & Asymptotes

In polar form, which equation represents a circle of radius 3 centred at the origin? Show your complete algebraic work.

Final Answer: $r = 3$. r = 3 means every point is exactly 3 units from the origin — a circle of radius 3.

16 [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 1.03 Rates of Change & Concavity

A particle's position in the Cartesian plane is described by the polar coordinates (r, θ), where r is the distance from the origin and θ is the angle measured counterclockwise from the positive x-axis. The particle begins at $\theta_1 = \frac{\pi}{6}$ with a radial distance of $r_1 = 4$. It then moves to a new position such that its new angle is $\theta_2 = \frac{2\pi}{3}$ and the radial distance is $r_2 = 6$. If the displacement vector of the particle is $\mathbf{d} = \langle x, y \rangle$, what is the value of $x^2 + y^2$? Show your complete algebraic work.

Final Answer: 49. The initial position of the particle in Cartesian coordinates is $(x_1, y_1) = (r_1\cos\theta_1, r_1\sin\theta_1)$. Given $r_1 = 4$ and $\theta_1 = \frac{\pi}{6}$, we have: $x_1 = 4\cos(\frac{\pi}{6}) = 4 \cdot \frac{\sqrt{3}}{2} = 2\sqrt{3}$ and $y_1 = 4\sin(\frac{\pi}{6}) = 4 \cdot \frac{1}{2} = 2$. So, the initial position is $(2\sqrt{3}, 2)$. The final position of the particle in Cartesian coordinates is $(x_2, y_2) = (r_2\cos\theta_2, r_2\sin\theta_2)$. Given $r_2 = 6$ and $\theta_2 = \frac{2\pi}{3}$, we have: $x_2 = 6\cos(\frac{2\pi}{3}) = 6 \cdot (-\frac{1}{2}) = -3$ and $y_2 = 6\sin(\frac{2\pi}{3}) = 6 \cdot \frac{\sqrt{3}}{2} = 3\sqrt{3}$. So, the final position is $(-3, 3\sqrt{3})$. The displacement vector is $\mathbf{d} = \langle x, y \rangle = \langle x_2 - x_1, y_2 - y_1 \rangle$. Therefore, $x = -3 - 2\sqrt{3}$ and $y = 3\sqrt{3} - 2$. We need to find $x^2 + y^2$, which is the square of the magnitude of the displacement vector. This is equivalent to the square of the distance between the initial and final points. $x^2 = (-3 - 2\sqrt{3})^2 = (-1)^2(3 + 2\sqrt{3})^2 = 9 + 12\sqrt{3} + (2\sqrt{3})^2 = 9 + 12\sqrt{3} + 12 = 21 + 12\sqrt{3}$ $y^2 = (3\sqrt{3} - 2)^2 = (3\sqrt{3})^2 - 2(3\sqrt{3})(2) + 2^2 = 27 - 12\sqrt{3} + 4 = 31 - 12\sqrt{3}$ Now, $x^2 + y^2 = (21 + 12\sqrt{3}) + (31 - 12\sqrt{3}) = 21 + 31 = 52$.

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17 [Challenging] • Target: Radian Measure & Unit Circle Coordinates • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Consider a particle whose position in the complex plane at time t (in seconds) is given by $z(t) = 3e^{i(\frac{\pi}{6}t + \frac{\pi}{3})}$. (a) Find the initial position of the particle at $t = 0$ in polar coordinates (r, θ), where $r \geq 0$ and $0 \leq \theta < 2\pi$. (b) Determine the angular velocity of the particle in radians per second. (c) If the particle starts at a point A and moves along a circular path, find the arc length traveled by the particle from $t = 0$ to $t = 4$ seconds.

Final Answer: (a) $(3, \frac{2\pi}{3})$, (b) $\frac{\pi}{6}$ rad/s, (c) $\frac{2\pi}{3}$. The position of the particle is given by $z(t) = 3e^{i(\frac{\pi}{6}t + \frac{\pi}{3})}$. This is in the polar form $z(t) = re^{i\theta(t)}$, where $r = 3$ and $\theta(t) = \frac{\pi}{6}t + \frac{\pi}{3}$. (a) The initial position at $t = 0$ is $z(0) = 3e^{i(\frac{\pi}{6}(0) + \frac{\pi}{3})} = 3e^{i\frac{\pi}{3}}$. In polar coordinates (r, θ), the radius is $r = 3$. The angle is $\theta(0) = \frac{\pi}{3}$. Since $0 \leq \frac{\pi}{3} < 2\pi$, the initial position is $(3, \frac{\pi}{3})$. (b) The angular velocity is the rate of change of the angle with respect to time. The angle is given by $\theta(t) = \frac{\pi}{6}t + \frac{\pi}{3}$. The angular velocity ω is $\frac{d\theta}{dt} = \frac{d}{dt}(\frac{\pi}{6}t + \frac{\pi}{3}) = \frac{\pi}{6}$ radians per second. (c) The particle moves along a circular path with radius $r = 3$. The angular velocity is $\omega = \frac{\pi}{6}$ rad/s. The time interval is $\Delta t = 4 - 0 = 4$ seconds. The total angle swept in this interval is $\Delta\theta = \omega\Delta t = \frac{\pi}{6} \times 4 = \frac{4\pi}{6} = \frac{2\pi}{3}$ radians. The arc length s is given by $s = r\Delta\theta$. So, $s = 3 \times \frac{2\pi}{3} = 2\pi$ units. Therefore, the correct choice is (a) $(3, \frac{\pi}{3})$, (b) $\frac{\pi}{6}$ rad/s, (c) 2π.

18 [Challenging] • Target: Exact Values of Trigonometric Functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

A point P has polar coordinates (r, θ). When converted to Cartesian coordinates, P is $(3, 3\sqrt{3})$. If the angle θ is restricted to the interval $[0, 2\pi)$, find the value of r and θ in radians.

Final Answer: $r = 6, \theta = \frac{\pi}{3}$. The conversion formulas from polar coordinates (r, θ) to Cartesian coordinates (x, y) are $x = r\cos\theta$ and $y = r\sin\theta$. We are given the Cartesian coordinates $x = 3$ and $y = 3\sqrt{3}$. We can find r using the relationship $r^2 = x^2 + y^2$. $r^2 = 3^2 + (3\sqrt{3})^2$ $r^2 = 9 + (9 \times 3)$ $r^2 = 9 + 27$ $r^2 = 36$ Since r represents a distance, it must be non-negative. Thus, $r = \sqrt{36} = 6$. Now we find θ using the relationships $\tan\theta = \frac{y}{x}$ (if $x \neq 0$) and considering the quadrant. $\tan\theta = \frac{3\sqrt{3}}{3} = \sqrt{3}$. The principal value for $\arctan(\sqrt{3})$ is $\frac{\pi}{3}$. Since $x = 3$ (positive) and $y = 3\sqrt{3}$ (positive), the point P is in the first quadrant. Therefore, $\theta = \frac{\pi}{3}$ is the correct angle within the interval $[0, 2\pi)$. Thus, the polar coordinates are $(6, \frac{\pi}{3})$.

19 [Challenging] • Target: Sinusoidal Graphs & Transformations • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Write a sinusoidal function with amplitude 5, period 4π, midline $y = -3$, using $\cos$.

Final Answer: $f(x) = 5\cos(1/2x) - 3$. Amplitude $= A = 5$; period $= 2\pi/B = 4\pi \Rightarrow B = 1/2$; midline $D = -3$. So $f(x) = 5\cos(1/2x) - 3$.

20 [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 4.01 Inverse & Composite Functions

Write a sinusoidal function with amplitude 5, period 3π, midline $y = -2$, using $\sin$.

Final Answer: $f(x) = 5\sin(2/3x) - 2$. Amplitude $= A = 5$; period $= 2\pi/B = 3\pi \Rightarrow B = 2/3$; midline $D = -2$. So $f(x) = 5\sin(2/3x) - 2$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Radian Measure & Unit Circle Coordinates

What are the exact coordinates of the terminal point on the unit circle for angle $\theta = \frac{5\pi}{6}$?

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

Angle $\frac{5\pi}{6}$ lies in Quadrant II with reference angle $\frac{\pi}{6}$. Therefore, $\cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$ and $\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$, giving coordinates $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

22 [Foundational] • Target: Radian Measure & Unit Circle Coordinates

Convert an angle of $\frac{7\pi}{4}$ radians into degree measure.

a. 315° ✓

b. 225°

c. 300°

d. 135°

Multiply by $\frac{180^\circ}{\pi}$: $\frac{7\pi}{4} \times \frac{180^\circ}{\pi} = 7 \times 45^\circ = 315^\circ$.

23 [Foundational] • Target: Radian Measure & Unit Circle Coordinates

If the terminal ray of angle α passes through the point $(-3, -4)$, what is $\sin\alpha$?

a. $-\frac{4}{5}$ ✓

b. $-\frac{3}{5}$

c. $\frac{4}{3}$

d. $\frac{3}{5}$

Radius $r = \sqrt{(-3)^2 + (-4)^2} = \sqrt{9 + 16} = 5$. Then $\sin\alpha = \frac{y}{r} = -\frac{4}{5}$.

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24 [Foundational] • Target: Exact Values of Trigonometric Functions

Evaluate $\tan\left(\frac{4\pi}{3}\right)$.

a. $\sqrt{3}$ ✓

b. $-\sqrt{3}$

c. $\frac{\sqrt{3}}{3}$

d. $-\frac{\sqrt{3}}{3}$

Angle $\frac{4\pi}{3}$ is in Quadrant III, where tangent is positive. The reference angle is $\frac{\pi}{3}$. $\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$, so $\tan\left(\frac{4\pi}{3}\right) = \sqrt{3}$.

25 [Foundational] • Target: Exact Values of Trigonometric Functions

What is the exact value of $\sec\left(-\frac{\pi}{4}\right)$?

a. $\sqrt{2}$ ✓

b. $-\sqrt{2}$

c. $\frac{\sqrt{2}}{2}$

d. 2

Cosine is an even function: $\cos(-\pi/4) = \cos(\pi/4) = \frac{\sqrt{2}}{2}$. Thus $\sec(-\pi/4) = \frac{1}{\cos(\pi/4)} = \frac{2}{\sqrt{2}} = \sqrt{2}$.

26 [Foundational] • Target: Exact Values of Trigonometric Functions

If $\sin\theta = \frac{3}{5}$ and $\cos\theta < 0$, find $\cot\theta$.

a. $-\frac{4}{3}$ ✓

b. $\frac{4}{3}$

c. $-\frac{3}{4}$

d. $-\frac{4}{5}$

Since $\sin\theta > 0$ and $\cos\theta < 0$, θ is in Quadrant II. Here $x = -\sqrt{5^2 - 3^2} = -4$. Then $\cot\theta = \frac{x}{y} = -\frac{4}{3}$.

27 [Foundational] • Target: Sinusoidal Graphs & Transformations

Find the amplitude and period of the sinusoidal function $f(x) = -4\cos(3x - \pi) + 2$.

a. **Amplitude** $= 4$**, Period** $= \frac{2\pi}{3}$ ✓

b. Amplitude $= -4$, Period $= 3$

c. Amplitude $= 4$, Period $= 3\pi$

d. Amplitude $= 2$, Period $= \frac{2\pi}{3}$

Amplitude is $|A| = |-4| = 4$. For $B = 3$, the period is $\frac{2\pi}{B} = \frac{2\pi}{3}$.

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28 [Foundational] • Target: Sinusoidal Graphs & Transformations

A Ferris wheel of radius 25 m completes one rotation every 60 seconds. Passengers board at the bottom, 2 m above the ground. Which cosine equation models height $h(t)$ in meters?

a. $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$ ✓

b. $h(t) = 25\cos\left(\frac{\pi}{30}t\right) + 27$

c. $h(t) = -25\cos\left(\frac{\pi}{60}t\right) + 2$

d. $h(t) = -27\cos\left(\frac{\pi}{30}t\right) + 25$

Centerline is $d = 2 + 25 = 27$. Amplitude is 25. Period is $60 \Rightarrow B = \frac{2\pi}{60} = \frac{\pi}{30}$. Since it starts at minimum height 2, an inverted cosine is used: $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Sinusoidal Graphs & Transformations

What is the phase shift of $g(x) = 2\sin\left(4x + \frac{\pi}{2}\right)$?

a. $\frac{\pi}{8}$ **units to the left** ✓

b. $\frac{\pi}{8}$ units to the right

c. $\frac{\pi}{2}$ units to the left

d. $\frac{\pi}{4}$ units to the left

Factor out 4: $g(x) = 2\sin\left(4\left(x + \frac{\pi}{8}\right)\right)$. The phase shift is $\frac{\pi}{8}$ units to the left.

30 [Intermediate] • Target: Fundamental Trigonometric Identities

Simplify the trigonometric expression $\frac{\tan x}{\sec x}$.

a. $\sin x$ ✓

b. $\cos x$

c. $\csc x$

d. $\cot x$

Rewrite in terms of sine and cosine: $\frac{\sin x/\cos x}{1/\cos x} = \frac{\sin x}{\cos x} \cdot \cos x = \sin x$.

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31 [Intermediate] • Target: Fundamental Trigonometric Identities

Which expression is equivalent to $(1-\cos^2\theta)(1+\cot^2\theta)$ for all defined θ?

a. 1 ✓

b. $\sin^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

Use Pythagorean identities: $1-\cos^2\theta=\sin^2\theta$ and $1+\cot^2\theta=\csc^2\theta$. Therefore, $\sin^2\theta\cdot\csc^2\theta=\sin^2\theta\cdot\frac{1}{\sin^2\theta}=1$.

32 [Intermediate] • Target: Fundamental Trigonometric Identities

Solve for $\theta\in[0,2\pi)$: $2\cos^2\theta-\cos\theta-1=0$.

a. $\theta=0,\frac{2\pi}{3},\frac{4\pi}{3}$ ✓

b. $\theta=\pi,\frac{\pi}{3},\frac{5\pi}{3}$

c. $\theta=0,\frac{\pi}{3},\frac{5\pi}{3}$

d. $\theta=\frac{\pi}{2},\frac{3\pi}{2}$

Factor: $(2\cos\theta+1)(\cos\theta-1)=0$. So $\cos\theta=1\Rightarrow\theta=0$, and $\cos\theta=-\frac{1}{2}\Rightarrow\theta=\frac{2\pi}{3},\frac{4\pi}{3}$.

33 [Intermediate] • Target: Radian Measure & Unit Circle Coordinates

Write a sinusoidal function with amplitude 5, period 3π, midline $y=-2$, using $\cos$.

a. $f(x)=5\cos(2/3x)-1$

b. $f(x)=6\cos(2/3x)-2$

c. $f(x)=5\cos(4/3x)-2$

d. $f(x)=5\cos(2/3x)-2$ ✓

Amplitude $=A=5$; period $=2\pi/B=3\pi\Rightarrow B=2/3$; midline $D=-2$. So $f(x)=5\cos(2/3x)-2$.

34 [Intermediate] • Target: Exact Values of Trigonometric Functions

Write a sinusoidal function with amplitude 5, period 4π, midline $y=-1$, using $\cos$.

a. $f(x)=5\cos(1/2x)-1$ ✓

b. $f(x)=5\cos(1x)-1$

c. $f(x)=6\cos(1/2x)-1$

d. $f(x)=5\cos(1/2x)+0$

Amplitude $=A=5$; period $=2\pi/B=4\pi\Rightarrow B=1/2$; midline $D=-1$. So $f(x)=5\cos(1/2x)-1$.

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35 [Intermediate] • Target: Sinusoidal Graphs & Transformations

Write a sinusoidal function with amplitude 3, period 4π, midline $y = 2$, using $\sin$.

a. $f(x) = 4\sin(1/2x) + 2$

b. $f(x) = 3\sin(1/2x) + 3$

c. $f(x) = 3\sin(1/2x) + 2$ ✓

d. $f(x) = 3\sin(1x) + 2$

Amplitude $= A = 3$; period $= 2\pi/B = 4\pi \Rightarrow B = 1/2$; midline $D = 2$. So $f(x) = 3\sin(1/2x) + 2$.

36 [Intermediate] • Target: Fundamental Trigonometric Identities

Write a sinusoidal function with amplitude 3, period 3π, midline $y = 1$, using $\sin$.

a. $f(x) = 3\sin(2/3x) + 1$ ✓

b. $f(x) = 3\sin(4/3x) + 1$

c. $f(x) = 3\sin(2/3x) + 2$

d. $f(x) = 4\sin(2/3x) + 1$

Amplitude $= A = 3$; period $= 2\pi/B = 3\pi \Rightarrow B = 2/3$; midline $D = 1$. So $f(x) = 3\sin(2/3x) + 1$.

37 [Intermediate] • Target: Radian Measure & Unit Circle Coordinates

Write a sinusoidal function with amplitude 2, period 3π, midline $y = 0$, using $\cos$.

a. $f(x) = 3\cos(2/3x)$

b. $f(x) = 2\cos(2/3x) + 1$

c. $f(x) = 2\cos(4/3x)$

d. $f(x) = 2\cos(2/3x)$ ✓

Amplitude $= A = 2$; period $= 2\pi/B = 3\pi \Rightarrow B = 2/3$; midline $D = 0$. So $f(x) = 2\cos(2/3x)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Exact Values of Trigonometric Functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Write a sinusoidal function with amplitude 5, period 2π, midline $y = -2$, using $\cos$.

a. $f(x) = 5\cos(2x) - 2$

b. $f(x) = 5\cos(1x) - 2$ ✓

c. $f(x) = 5\cos(1x) - 1$

d. $f(x) = 6\cos(1x) - 2$

Amplitude $= A = 5$; period $= 2\pi/B = 2\pi \Rightarrow B = 1$; midline $D = -2$. So $f(x) = 5\cos(1x) - 2$.

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39 [Challenging] • Target: Sinusoidal Graphs & Transformations • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Write a sinusoidal function with amplitude 4, period 2π, midline $y = -3$, using $\cos$.

a. $f(x) = 4\cos(1x) - 2$

b. $f(x) = 4\cos(2x) - 3$

c. $f(x) = 5\cos(1x) - 3$

d. $f(x) = 4\cos(1x) - 3$ ✓

Amplitude $= A = 4$; period $= 2\pi/B = 2\pi \Rightarrow B = 1$; midline $D = -3$. So $f(x) = 4\cos(1x) - 3$.

40 [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 4.01 Inverse & Composite Functions

Write a sinusoidal function with amplitude 2, period 2π, midline $y = 3$, using $\sin$.

a. $f(x) = 2\sin(2x) + 3$

b. $f(x) = 2\sin(1x) + 4$

c. $f(x) = 2\sin(1x) + 3$ ✓

d. $f(x) = 3\sin(1x) + 3$

Amplitude $= A = 2$; period $= 2\pi/B = 2\pi \Rightarrow B = 1$; midline $D = 3$. So $f(x) = 2\sin(1x) + 3$.

41 [Challenging] • Target: Radian Measure & Unit Circle Coordinates • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Write a sinusoidal function with amplitude 3, period π, midline $y = -2$, using $\cos$.

a. $f(x) = 4\cos(2x) - 2$

b. $f(x) = 3\cos(2x) - 1$

c. $f(x) = 3\cos(4x) - 2$

d. $f(x) = 3\cos(2x) - 2$ ✓

Amplitude $= A = 3$; period $= 2\pi/B = \pi \Rightarrow B = 2$; midline $D = -2$. So $f(x) = 3\cos(2x) - 2$.

42 [Challenging] • Target: Exact Values of Trigonometric Functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

Write a sinusoidal function with amplitude 4, period π, midline $y = -3$, using $\cos$.

a. $f(x) = 4\cos(2x) - 3$ ✓

b. $f(x) = 4\cos(4x) - 3$

c. $f(x) = 5\cos(2x) - 3$

d. $f(x) = 4\cos(2x) - 2$

Amplitude $= A = 4$; period $= 2\pi/B = \pi \Rightarrow B = 2$; midline $D = -3$. So $f(x) = 4\cos(2x) - 3$.

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43 [Challenging] • Target: Sinusoidal Graphs & Transformations • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Write a sinusoidal function with amplitude 4, period π, midline $y = -3$, using $\sin$.

a. $f(x) = 4\sin(4x) - 3$

b. $f(x) = 4\sin(2x) - 2$

c. $f(x) = 5\sin(2x) - 3$

d. $f(x) = 4\sin(2x) - 3$ ✓

Amplitude $= A = 4$; period $= 2\pi/B = \pi \Rightarrow B = 2$; midline $D = -3$. So $f(x) = 4\sin(2x) - 3$.

44 [Challenging] • Target: Fundamental Trigonometric Identities • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Write a sinusoidal function with amplitude 5, period π, midline $y = -1$, using $\sin$.

a. $f(x) = 5\sin(2x) + 0$

b. $f(x) = 6\sin(2x) - 1$

c. $f(x) = 5\sin(4x) - 1$

d. $f(x) = 5\sin(2x) - 1$ ✓

Amplitude $= A = 5$; period $= 2\pi/B = \pi \Rightarrow B = 2$; midline $D = -1$. So $f(x) = 5\sin(2x) - 1$.

45 [Challenging] • Target: Radian Measure & Unit Circle Coordinates • Cross-Lesson: 1.03 Rates of Change & Concavity

Write a sinusoidal function with amplitude 4, period π, midline $y = 2$, using $\sin$.

a. $f(x) = 5\sin(2x) + 2$

b. $f(x) = 4\sin(4x) + 2$

c. $f(x) = 4\sin(2x) + 2$ ✓

d. $f(x) = 4\sin(2x) + 3$

Amplitude $= A = 4$; period $= 2\pi/B = \pi \Rightarrow B = 2$; midline $D = 2$. So $f(x) = 4\sin(2x) + 2$.

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